

1 Latent Representation Probing for Toxic Comment Detection

1.1 Impact Statement Card

1.1.1 Positive Social and Safety Impact

Community safety. A toxicity detector can support real-time screening by flagging potentially harmful content before publication or prioritizing it for review. Used carefully, this can help foster healthier online discussions.

Moderator wellbeing. Automated first-pass screening can reduce the volume of extreme or repetitive content that human moderators must inspect, while preserving human judgment for ambiguous cases.

Computational efficiency. A frozen language model plus a lightweight probe may require less training and inference overhead than end-to-end fine-tuning or full generative inference for every comment. The actual savings depend on model size, hardware, batch size, and deployment architecture and should be measured rather than assumed.

1.1.2 Potential Risks and Mitigation Strategies

1.1.2.1 Risk 1: False positives and over-moderation

A detector may suppress legitimate speech, reclaimed language, political discussion, or benign slang.

Mitigation: Use calibrated probability thresholds; reserve automatic removal for high-confidence cases; route lower-confidence decisions to human review; monitor appeal and override rates; and evaluate across relevant dialects and communities.

1.1.2.2 Risk 2: False negatives and under-moderation

Harmful content may evade detection, particularly when it is oblique, coded, multilingual, or outside the training distribution.

Mitigation: Evaluate on adversarial and out-of-distribution test sets, refresh the evaluation suite, conduct targeted error analysis, and combine model scores with reporting and moderation workflows.

1.1.2.3 Risk 3: Feedback-loop effects

If model decisions influence which examples are later collected, the training data may overrepresent previously detected patterns and underrepresent missed harms.

Mitigation: Preserve random audit samples, track disagreement and appeals, and periodically sample content independently of model confidence.

1.1.3 Environmental and Operational Impact

Latent probing can avoid end-to-end updates to the base model, potentially reducing training compute and operational resource requirements. The environmental impact should be assessed with measured energy, hardware, and latency data across the intended deployment setting.

1.1.4 Responsible Deployment Principles

The system should provide scores and evidence for review rather than presenting uncertain classifications as facts. Deployment should include threshold documentation, access controls, audit logs, user appeal paths, privacy safeguards, and a process for withdrawing or recalibrating the model when harms are observed.

Prepared from the supplied project outline; no unverified performance or energy figures are asserted.